INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Meta Sarpy Data center, NE -- station KMLE, America/Chicago
Committed pair	OSM 844394339 -> 942557184 Meta Sarpy Data center / Travelers Insurance
Facade gap	262.7 m between the two halls
Weather record	43,165 real hourly records from KMLE
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3794 C at 155 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-03-30 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 21 hours with 2 mode changes. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 21 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 21 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	13.332	18.0	14.000	4.644
01	FREE	yes	-	13.509	18.0	13.330	4.090
02	FREE	yes	-	13.888	18.0	11.220	3.704
03	FREE	yes	-	14.058	18.0	8.780	3.286
04	FREE	yes	-	14.292	18.0	8.000	3.159
05	FREE	yes	-	14.367	18.0	7.220	2.940
06	FREE	yes	-	14.708	18.0	5.890	2.804
07	FREE	yes	-	15.343	18.0	5.280	3.042

08	FREE	yes	-	15.691	18.0	6.220	3.585
09	FREE	yes	-	16.354	18.0	7.220	4.683
11	mechanical	no	dry-bulb	20.402	18.0	9.000	6.031
12	mechanical	no	dry-bulb	21.267	18.0	9.440	6.512
13	mechanical	no	dry-bulb	20.183	18.0	10.560	6.556
15	mechanical	yes	switch budget	15.479	18.0	10.110	4.678
16	mechanical	yes	switch budget	14.156	18.0	10.720	4.108
17	mechanical	yes	switch budget	12.954	18.0	9.890	3.788
18	mechanical	yes	switch budget	12.232	18.0	9.220	3.368
19	mechanical	yes	switch budget	10.578	18.0	8.110	3.291
20	mechanical	yes	switch budget	10.194	18.0	7.500	3.592
21	mechanical	yes	switch budget	9.209	18.0	5.890	4.289
23	mechanical	yes	switch budget	10.221	18.0	2.619	4.982

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.332 C, which is 4.668 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.644 C, and the margin is measured, not chosen: 4.423 C of group-conditional forecast error for this hour of day, plus 0.2215 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.510 C, which is 4.490 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.090 C, and the margin is measured, not chosen: 3.860 C of group-conditional forecast error for this hour of day, plus 0.2295 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.888 C, which is 4.112 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.704 C, and the margin is measured, not chosen: 3.487 C of group-conditional forecast error for this hour of day, plus 0.2177 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.058 C, which is 3.942 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.286 C, and the margin is measured, not chosen: 3.059 C of group-conditional forecast error for this hour of day, plus 0.2276 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.292 C, which is 3.708 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.159 C, and the margin is measured, not chosen: 2.867 C of group-conditional forecast error for this hour of day, plus 0.2921 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.367 C, which is 3.633 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.940 C, and the margin is measured, not chosen: 2.683 C of group-conditional forecast error for this hour of day, plus 0.2566 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.708 C, which is 3.292 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.804 C, and the margin is measured, not chosen: 2.596 C of group-conditional forecast error for this hour of day, plus 0.2079 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.343 C, which is 2.657 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.042 C, and the margin is measured, not chosen: 2.809 C of group-conditional forecast error for this hour of day, plus 0.2333 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.691 C, which is 2.309 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.585 C, and the margin is measured, not chosen: 3.334 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.354 C, which is 1.646 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.683 C, and the margin is measured, not chosen: 4.439 C of group-conditional forecast error for this hour of day, plus 0.2441 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.402 C against a 18.0 C limit, so it fails by 2.402 C. It would take a limit 2.402 C higher, or a bound 2.402 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.267 C against a 18.0 C limit, so it fails by 3.267 C. It would take a limit 3.267 C higher, or a bound 3.267 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.183 C against a 18.0 C limit, so it fails by 2.183 C. It would take a limit 2.183 C higher, or a bound 2.183 C tighter, to change this hour.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.479 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.156 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.954 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.232 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.578 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.195 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.209 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.221 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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